

Disproof of TLMC Conjecture 00000000521 (v2: Betti numbers of the ideal I , literal reading)

TLMC falsification submission

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Abstract

We disprove TLMC conjecture 00000000521 *under its literal reading*. The conjecture defines the alternating Betti sum $A_j = \sum_i (-1)^i \beta_{i,j}(I)$ using the graded Betti numbers *of the ideal I as a graded R -module* ($\text{Tor}^R(I, k)$), where $\beta_{0,j}(I)$ counts the minimal generators of I . The 4-cycle C_4 with $I = (x_1x_2, x_2x_3, x_3x_4, x_4x_1) \subset k[x_1, x_2, x_3, x_4]$ is a counterexample: resolving the *module I* , one gets $\beta_{0,2}(I) = 4$, $\beta_{1,3}(I) = 4$, $\beta_{2,4}(I) = 1$, so $\text{reg}(I) = 2$ and $A = (A_0, \dots, A_4) = (0, 0, 4, -4, 1)$, whose signs $+, -, +$ change *twice*, not exactly once. The monotonicity clause holds for this example, so the failure is exactly the “exactly once” clause; the star $K_{1,3}$ ($A = (0, 0, 3, -3, 1)$) is a second counterexample. A v1 submission used the Betti table of R/I instead of that of I ; all data here are recomputed for the module I by four independent methods and formalized in axiom-free Lean 4.

1 The conjecture and its object

The conjecture text reads, verbatim:

“Definition: The alternating Betti sum $A_j = \sum_i (-1)^i \beta_{i,j}(I)$ of a monomial ideal I .
Conjecture: For all edge ideals, the sign of $\{A_j\}$ changes exactly once as j crosses $\text{reg}(I)$, and $|A_j|$ is monotonically nondecreasing up to that point; this gives a combinatorial characterization of the graded Lefschetz property for edge ideals.”

The object is fixed by the notation $\beta_{i,j}(I)$: the graded Betti numbers *of the ideal I as a graded R -module*, i.e. $\beta_{i,j}(I) = \dim_k \text{Tor}_i^R(I, k)_j$. Thus $\beta_{0,j}(I)$ counts the minimal generators of I in degree j , and $\beta_{0,0}(I) = 0$ for every nonzero ideal. A v1 submission instead used the Betti table of R/I (which starts with $\beta_{0,0} = 1$); that version was closed after review, and we thank the reviewer for the correction. Everything below is computed for the module I .

Remark 1 (Relating the two readings). *For an edge ideal $I = I(G) \subset R = k[x_1, \dots, x_n]$, the long exact Tor sequence of $0 \rightarrow I \rightarrow R \rightarrow R/I \rightarrow 0$ gives $\beta_{i,j}(I) = \beta_{i+1,j}(R/I)$ for $i \geq 1$ and $\beta_{0,j}(I) = \beta_{1,j}(R/I)$ (as $I/\mathfrak{m}I$ is concentrated in degree 2). Consequently*

$$A_j(I) = \delta_{j0} - A_j(R/I), \tag{1}$$

i.e. the literal sequence is the R/I sequence with its leading $A_0(R/I) = 1$ removed and negated. In particular the literal conjecture holds for G iff the R/I -sequence $\{A_j(R/I)\}_{j \geq 1}$ changes sign exactly twice (at the right place); any graph with $\beta_{3,4}(R/I) \neq 0$ produces two changes literally.

2 The counterexample C_4

Let $G = C_4$ be the 4-cycle and $I = I(C_4) = (x_1x_2, x_2x_3, x_3x_4, x_4x_1) \subset R = k[x_1, x_2, x_3, x_4]$. The minimal free resolution of the module I is

$$0 \longrightarrow R(-4) \xrightarrow{d_2} R(-3)^4 \xrightarrow{d_1} R(-2)^4 \longrightarrow I \longrightarrow 0, \quad (2)$$

where d_1 is given by the four adjacent-edge syzygies

$$s_{12} = (x_3, -x_1, 0, 0), \quad s_{23} = (0, x_4, -x_2, 0), \quad s_{34} = (0, 0, x_1, -x_3), \quad s_{41} = (x_4, 0, 0, -x_2),$$

and d_2 by their single relation $\rho = (x_4, x_1, x_2, -x_3)$: $x_4s_{12} + x_1s_{23} + x_2s_{34} - x_3s_{41} = 0$. (The opposite-pair syzygies are generated by these: e.g. $(x_3x_4, 0, -x_1x_2, 0) = x_4s_{12} + x_1s_{23}$; the coefficient matrix of $(s_{12}, s_{23}, s_{34}, s_{41})$ has rank 3 and determinant 0, and its kernel over R is exactly $R \cdot \rho$, since $\gcd(x_1, x_2, x_3, x_4) = 1$.) All entries of d_1, d_2 have degree 1, so the resolution is minimal. Exactness and minimality of (2) are verified mechanically in `reproduce.py`: $d_1d_2 = 0$, per-bidegree homology $H_1 = H_2 = 0$ over $\text{GF}(10^9+7)$, and $\text{coker}(d_1)$ matches the Hilbert series of I degree by degree. Hence

$$\beta_{0,2}(I) = 4, \quad \beta_{1,3}(I) = 4, \quad \beta_{2,4}(I) = 1, \quad \beta_{i,j}(I) = 0 \text{ else,}$$

so $\text{reg}(I) = \max(j - i) = 2$, and

$$A_0 = 0, \quad A_1 = 0, \quad A_2 = +4, \quad A_3 = -4, \quad A_4 = +1, \quad A_j = 0 \ (j \geq 5).$$

Theorem 2 (Falsification, literal reading). *Conjecture 00000000521 is false. For the edge ideal $I = (x_1x_2, x_2x_3, x_3x_4, x_4x_1)$ of the 4-cycle, the nonzero entries of $\{A_j\}_{j \geq 0}$ are 4, -4, 1 with signs +, -, +: the sign changes twice, not exactly once. The first change does occur as j crosses $\text{reg}(I) = 2$ ($A_2 = 4 > 0 > A_3 = -4$), but a second change follows at $j = 3$ ($-4 \rightarrow +1$). Meanwhile $|A_0| \leq |A_1| \leq |A_2|$, so the monotonicity clause holds: the counterexample refutes exactly the “exactly once” clause.*

Proof. The Betti numbers displayed after (2) are those of the module I (exact minimal resolution, verified as described). Substituting into $A_j = \sum_i (-1)^i \beta_{i,j}(I)$ gives $A = (0, 0, 4, -4, 1)$; the sign scan is arithmetic. All steps are additionally cross-checked by the Hilbert-series identity $A_j = [t^j](1-t)^4 \text{Hilb}_I(t)$ and by (1), and are formalized in axiom-free Lean 4 (see Section 4). $\square$

Remark 3 (The $v1$ example is not a counterexample, literally). *For the triangle C_3 , $I = (xy, xz, yz)$ resolves as $0 \rightarrow R(-3)^2 \rightarrow R(-2)^3 \rightarrow I \rightarrow 0$, giving $A(I) = (0, 0, 3, -2)$: exactly one change at $\text{reg}(I) = 2$. The conjecture holds for C_3 under the literal reading, as the reviewer observed; likewise for P_3 and P_4 (same sequence) and for $2K_2$ ($A = (0, 0, 2, 0, -1)$, one change at $\text{reg} = 3$). The $v1$ numbers $A = (1, 0, -3, 2)$ came from the R/I table.*

3 Boundary: how widespread the literal failure is

All computations come from `reproduce.py` (module- I Taylor resolution $\otimes k$, Hilbert-cross-checked and identity-checked via (1)); every example and every graph in the scans passed.

graph G	$A_j(I) = \sum_i (-1)^i \beta_{i,j}(I)$	$\text{reg}(I)$	changes
single edge K_2	$(0, 0, 1)$	2	0
P_3, P_4, C_3	$(0, 0, 3, -2)$	2	1 (holds)
$2K_2$	$(0, 0, 2, 0, -1)$	3	1 (holds)
star $K_{1,3}$	$(0, 0, 3, -3, 1)$	2	2
C_4	$(0, 0, 4, -4, 1)$	2	2
paw	$(0, 0, 4, -4, 1)$	2	2
diamond	$(0, 0, 5, -6, 2)$	2	2
K_4	$(0, 0, 6, -8, 3)$	2	2
C_5	$(0, 0, 5, -5, 0, 1)$	3	2
C_6	$(0, 0, 6, -6, -3, 6, -2)$	3	3
$P_3 \sqcup K_2$	$(0, 0, 3, -1, -2, 1)$	3	2, and $ A_2 =3 > A_3 =1$

Notes.

- The single-edge case is degenerate ($I \cong R(-2)$ is free, so $A = (0, 0, 1)$ never changes sign at all) and is listed only for completeness; the counterexamples above are non-degenerate.
- Exhaustive scan, literal reading: among the 63 labeled graphs on 4 vertices with at least one edge, **32** violate “exactly one sign change”; among the 1023 on 5 vertices, **908** do, and **100** additionally violate the monotonicity clause (e.g. $P_3 \sqcup K_2$ in the table).
- By Remark 1, the literal failure class is exactly the class of graphs whose R/I -table changes sign three or more times, e.g. every graph with $\beta_{3,4}(R/I) \neq 0$ ($C_4, K_{1,3}, K_4, C_5, C_6$, paw, diamond, ...).

4 Verification

Independent recomputation (reproduce.py). The script (stdlib only) computes $\beta_{i,j}(I)$ for the module I as the homology of the Taylor resolution of I tensored with $k = \text{GF}(10^9+7)$ (Tor is computable from any free resolution; after tensoring, only the constant Taylor entries survive, and ranks are computed on $0, \pm 1$ matrices, so the output is exact). Four cross-checks run throughout: (i) A_j must equal $[t^j](1-t)^n \text{Hilb}_I(t)$ from direct monomial enumeration; (ii) the identity (1) against an independently computed R/I -table; (iii) $\beta_{0,2}^I = \#\{\text{edges}\}$ and $\beta_{0,0}^I = 0$; (iv) for C_4 , the displayed resolution (2) is checked entry-by-entry: $d_1 d_2 = 0$, per-bidegree homology $H_1 = H_2 = 0$, $\text{coker}(d_1)$ vs. Hilb_I , and minimality. Run: `python3 reproduce.py` (exit 0).

Lean 4 formalization (lean4/). Under the literal definition, the Betti table of the module $I(C_4)$ ($\beta_{0,2} = 4, \beta_{1,3} = 4, \beta_{2,4} = 1$), the alternating sums $A = (0, 0, 4, -4, 1)$, $A_j = 0$ for $j \geq 5$, $\text{reg}(I) = 2$ (attained and maximal), the *two* sign changes ($\neq 1$), the holding of the monotonicity clause, and the second counterexample $K_{1,3}$ are proved by `decide/rfl` in pure Lean 4 core (v4.33.1, no Mathlib). `lake build && lake env lean Check.lean` succeeds and every one of the 15 `#print axioms` reports “does not depend on any axioms”; there is no `sorry` and no additional axiom.

Changes from v1 (closed PR #130). Object corrected to the ideal I (literal reading); counterexample replaced: C_3 (which holds literally) by C_4 , with $K_{1,3}$ added; all tables, sums and scans recomputed for the module I ; Lean theorem now refutes the conjecture under its literal $\beta_{i,j}(I)$ definition, as requested by the reviewer.